

Safety data sheet

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BASF Safety data sheet according to Regulation (EC) No. 1907/2006 as amended from time to time.

Date / Revised: 03.01.2020

Version: 12.0

Date previous version: 04.08.2016

Previous version: 11.1

Product: **Sodium Nitrite food grade E250**

(ID no. 30216106/SDS_GEN_GB/EN)

Date of print 04.01.2020

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Sodium Nitrite food grade E250

Chemical name: sodium nitrite

INDEX-Number: 007-010-00-4

CAS Number: 7632-00-0

REACH registration number: 01-2119471836-27-0000

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: food additive(s)

Recommended use: Raw material, Intermediate, food additive(s), corrosion inhibitor, Surface treatment agent

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:

BASF SE
67056 Ludwigshafen
GERMANY

Contact address:

BASF plc
PO Box 4, Earl Road, Cheadle Hulme,
Cheadle, Cheshire
SK8 6QG, UNITED KINGDOM

Telephone: +44 161 485-6222

E-mail address: product-safety-north@basf.com

1.4. Emergency telephone number

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International emergency number:

Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

Ox. Sol. 3

Acute Tox. 3 (oral)

Eye Dam./Irrit. 2

Aquatic Acute 1

M-factor acute: 1

H319, H301, H400, H272

According to BASF current knowledge and application of the criteria given in Annex I of Regulation (EC) No. 1272/2008, the following classification exceeding the classification given in Regulation (EC) No 1272/2008, Annex VI, Table 3.1 is required.

Ox. Sol. 2

Acute Tox. 3 (oral)

Eye Dam./Irrit. 2

Aquatic Acute 1

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

Globally Harmonized System, EU (GHS)

Pictogram:



Signal Word:

Danger

Hazard Statement:

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H319	Causes serious eye irritation.
H301	Toxic if swallowed.
H400	Very toxic to aquatic life.
H272	May intensify fire; oxidizer.

Precautionary Statements (Prevention):

P273	Avoid release to the environment.
P280 + P283	Wear protective gloves and eye/face protection and fire/flame resistant/retardant clothing.
P280	Wear eye and/or face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.
P221	Take any precaution to avoid mixing with combustibles ...
P220	Keep away from clothing and other combustible materials.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or doctor/physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P301 + P330	IF SWALLOWED: rinse mouth.
P391	Collect spillage.
P337 + P311	If eye irritation persists: Call a POISON CENTER or doctor/physician.
P370 + P378	In case of fire: Use water spray for extinction.

Precautionary Statements (Storage):

P405	Store locked up.
P420	Store separately.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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According to Regulation (EC) No 1272/2008 [CLP]

Hazard determining component(s) for labelling: sodium nitrite

2.3. Other hazardsAccording to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

No specific dangers known, if the regulations/notes for storage and handling are considered.

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SECTION 3: Composition/Information on Ingredients

3.1. Substances

Chemical nature

sodium nitrite

CAS Number: 7632-00-0

EC-Number: 231-555-9

NaNO₂

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

3.2. Mixtures

Not applicable

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position).

If inhaled:

After inhalation of decomposition products, remove the affected person to a source of fresh air and keep calm. Provide medical aid. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:

Wash thoroughly with soap and water.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Overexposure may cause: vomiting, convulsions, cyanosis, death, coma, methaemoglobinaemia, nausea

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Hazards: Risk of pulmonary edema. Symptoms can appear later. Danger of methaemoglobin formation after ingestion.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), treat with toluonium chloride to reverse methaemoglobinanaemia.

SECTION 5: Fire-Fighting Measures**5.1. Extinguishing media**

Suitable extinguishing media:
water spray

Unsuitable extinguishing media for safety reasons:
ABC powder, carbon dioxide

5.2. Special hazards arising from the substance or mixture

nitrogen oxides

The substances/groups of substances mentioned can be released in case of fire. Has a fire-promoting effect due to release of oxygen.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:

Substance/product is an oxidizing agent and can supply oxygen to stimulate or accelerate the combustion of organic or other combustible substances/products.

SECTION 6: Accidental Release Measures**6.1. Personal precautions, protective equipment and emergency procedures**

Use breathing apparatus if exposed to vapours/dust/aerosol. Avoid contact with eyes.

6.2. Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into waterways or sewer systems without proper authorization.

6.3. Methods and material for containment and cleaning up

For residues: Pick up with suitable appliance and dispose of.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Keep container tightly sealed. Breathing must be protected when large quantities are decanted without local exhaust ventilation. Processing machines must be fitted with local exhaust ventilation. Protect against moisture. Protect against heat. Do not mix with combustible substances. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:

The substance/product is non-combustible. Has a fire-promoting effect due to release of oxygen. Where required Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

7.2. Conditions for safe storage, including any incompatibilities

Segregate from oxidizable substances. Segregate from acids. Segregate from ammonium salts.

Suitable materials for containers: Carbon steel (Iron), Stainless steel 1.4541, Stainless steel 1.4571, High density polyethylene (HDPE), Low density polyethylene (LDPE), rubberized
Further information on storage conditions: Keep container tightly closed and in a well-ventilated place. This product is classified as a dangerous substance for storage. The authority permits and storage regulations must be observed. Keep away from food, drink and animal feeding stuffs.

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

No occupational exposure limits known.

PNEC

freshwater: 0.0054 mg/l

marine water: 0.00616 mg/l

intermittent release: 0.0054 mg/l

sediment (freshwater): 0.0195 mg/kg

sediment (marine water): 0.0223 mg/kg

soil: 0.000733 mg/kg

STP: 21 mg/l

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DNEL

worker:

Long- and short-term exposure - systemic effects, Inhalation: 2 mg/m³**8.2. Exposure controls**Personal protective equipment

Respiratory protection:

Breathing protection if dusts are formed. Particle filter with high efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P3 or FFP3).

Hand protection:

Chemical resistant protective gloves (EN 374)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):

polyvinylchloride (PVC) - 0.7 mm coating thickness

nitrile rubber (NBR) - 0.4 mm coating thickness

chloroprene rubber (CR) - 0.5 mm coating thickness

butyl rubber (butyl) - 0.7 mm coating thickness

fluoroelastomer (FKM) - 0.7 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Do not breathe dust. Keep away from food, drink and animal feeding stuffs. No eating, drinking, smoking or tobacco use at the place of work. Take off immediately all contaminated clothing. Hands and/or face should be washed before breaks and at the end of the shift.

SECTION 9: Physical and Chemical Properties**9.1. Information on basic physical and chemical properties**

Form: crystalline

Colour: white to slightly yellow

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Odour:	faint odour
Odour threshold:	
pH value:	Not determined due to potential health hazard by inhalation. approx. 7 - 9 (100 g/l) The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.
Melting point:	280 °C
Boiling point:	(1,013.25 hPa) The substance / product decomposes therefore not determined.
Flash point:	Study scientifically not justified.
Evaporation rate:	
Flammability:	The product is a non-volatile solid. not highly flammable (other)
Lower explosion limit:	
Upper explosion limit:	For solids not relevant for classification and labelling.
Vapour pressure:	For solids not relevant for classification and labelling.
Density:	Study scientifically not justified. 2.17 g/cm ³ (20 °C) Literature data.
Relative density:	2.17 (20 °C) Literature data.
Solubility in water:	readily soluble
Partitioning coefficient n-octanol/water (log K _{ow}):	Study scientifically not justified.
Thermal decomposition:	> 320 °C Nitrogen monoxide, nitrogen dioxide, Disodium oxide
Viscosity, dynamic:	
Explosion hazard:	Study scientifically not justified. Based on the chemical structure there is no indicating of explosive properties.
Fire promoting properties:	Oxidizing.

9.2. Other information

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Self heating ability:	It is not a substance capable of spontaneous heating.	
Radioactivity:		not radioactive for transport purposes
Bulk density:	1,100 - 1,300 kg/m ³	
pKA:	Study scientifically not justified.	
Hygroscopy:	hygroscopic	
Surface tension:	Based on chemical structure, surface activity is not to be expected.	

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated. In the presence of water or moisture metal corrosion cannot be excluded.

10.2. Chemical stability

The product is chemically stable.

10.3. Possibility of hazardous reactions

Hazardous reactions in presence of mentioned substances to avoid.

The product is stable if stored and handled as prescribed/indicated. Reacts with organic substances.

10.4. Conditions to avoid

See SDS section 7 - Handling and storage.

10.5. Incompatible materials

Substances to avoid:

reducing agents, oxidizable substances, ammonium salts, amines, amine compounds, acids

10.6. Hazardous decomposition products

Hazardous decomposition products:

Disodium oxide

nitrogen oxides

SECTION 11: Toxicological Information

11.1. Information on toxicological effects

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Acute toxicity

Assessment of acute toxicity:

Of high toxicity after single ingestion. There is a risk of damage to the blood (methemoglobinemia) after a single uptake.

Experimental/calculated data:

LD50 rat (oral): 180 mg/kg

(by inhalation): Study scientifically not justified.

(dermal): Study scientifically not justified.

Irritation

Assessment of irritating effects:

Not irritating to the skin. Eye contact causes irritation.

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: Irritant. (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:

There is no evidence of a skin-sensitizing potential.

Experimental/calculated data:

Study scientifically not justified.

Germ cell mutagenicity

Assessment of mutagenicity:

The data available on mutagenic action are not consistent.

Carcinogenicity

Assessment of carcinogenicity:

In long-term studies in rats and mice in which the substance was given by drinking-water, a carcinogenic effect was not observed. Under certain conditions nitrites can enhance the formation of nitrosamines in vivo. Nitrosamines are carcinogenic in animal studies.

Reproductive toxicity

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:

In animal studies the substance did not cause malformations. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals. After the uptake of small doses toxicity to development will not be expected in humans.

Specific target organ toxicity (single exposure)Assessment of STOT single:

There is a risk of damage to the blood (methemoglobinemia) after a single uptake.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)Assessment of repeated dose toxicity:

After repeated administration the prominent effect is damage of the blood (methemoglobin formation).

Aspiration hazard

No aspiration hazard expected.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:

Very toxic (acute effect) to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:

LC50 (96 h) 0.54 - 26.3 mg/l, *Salmo gairdneri*, syn. *O. mykiss* (other, Flow through.)

Aquatic invertebrates:

LC50 (96 h) 4.93 mg/l, aquatic crustacea (static)

Literature data.

EC50 (48 h) 15.4 mg/l, *Daphnia magna* (OECD Guideline 202, part 1, static)

The statement of the toxic effect relates to the analytically determined concentration.

Aquatic plants:

EC50 (72 h) > 100 mg/l (growth rate), *Scenedesmus subspicatus* (OECD Guideline 201, static)

The statement of the toxic effect relates to the analytically determined concentration.

Microorganisms/Effect on activated sludge:

EC10 (3 h) 210 mg/l, activated sludge, domestic (OECD Guideline 209, static)

The details of the toxic effect relate to the nominal concentration.

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EC50 (48 h) 421 mg/l, protozoa (other, static)

Chronic toxicity to fish:

No observed effect concentration (31 d) 6.16 mg/l, *Ictalurus punctatus*, syn: *I. robustus* (other, Flow through.)

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (80 d) 9.86 mg/l, aquatic crustacea (*Daphnia* test chronic, static)

Assessment of terrestrial toxicity:

No data available.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Not applicable for inorganic substances.

Elimination information:

not applicable

Assessment of stability in water:

According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis):

not applicable

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:

Accumulation in organisms is not to be expected.

Bioaccumulation potential:

Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

12.4. Mobility in soil

Assessment transport between environmental compartments:

Volatility: The substance will not evaporate into the atmosphere from the water surface.

Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

12.6. Other adverse effects

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The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.7. Additional information

Other ecotoxicological advice:

Do not allow to enter soil, waterways or waste water channels. Do not release untreated into natural waters. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Contact manufacturer regarding recycling.

Check for possible recycling.

Contact waste centre regarding recycling.

The UK Environmental Protection (Duty of Care) Regulations (EP) and amendments should be noted (United Kingdom).

This product and any uncleaned containers must be disposed of as hazardous waste in accordance with the 2005 Hazardous Waste Regulations and amendments (United Kingdom)

Contaminated packaging:

Contaminated packaging should be emptied as far as possible and disposed of in accordance with official regulations after being thoroughly cleaned.

SECTION 14: Transport Information

Land transport

ADR

UN number	UN1500
UN proper shipping name:	SODIUM NITRITE
Transport hazard class(es):	5.1, 6.1, EHS
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	Tunnel code: E

RID

UN number	UN1500
UN proper shipping name:	SODIUM NITRITE

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Transport hazard class(es): 5.1, 6.1, EHS
Packing group: III
Environmental hazards: yes
Special precautions for user: None known

Inland waterway transport

ADN

UN number: UN1500
UN proper shipping name: SODIUM NITRITE
Transport hazard class(es): 5.1, 6.1, EHS
Packing group: III
Environmental hazards: yes
Special precautions for user: None known

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

UN number: UN 1500
UN proper shipping name: SODIUM NITRITE
Transport hazard class(es): 5.1, 6.1, EHS
Packing group: III
Environmental hazards: yes
Marine pollutant: YES
Special precautions for user: None known

Air transport

IATA/ICAO

UN number: UN 1500
UN proper shipping name: SODIUM NITRITE
Transport hazard class(es): 5.1, 6.1
Packing group: III
Environmental hazards: No Mark as dangerous for the environment is needed
Special precautions for user: None known

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14.1. UN number

See corresponding entries for "UN number" for the respective regulations in the tables above.

14.2. UN proper shipping name

See corresponding entries for "UN proper shipping name" for the respective regulations in the tables above.

14.3. Transport hazard class(es)

See corresponding entries for "Transport hazard class(es)" for the respective regulations in the tables above.

14.4. Packing group

See corresponding entries for "Packing group" for the respective regulations in the tables above.

14.5. Environmental hazards

See corresponding entries for "Environmental hazards" for the respective regulations in the tables above.

14.6. Special precautions for user

See corresponding entries for "Special precautions for user" for the respective regulations in the tables above.

14.7. Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

Further information

This product is subject to the most recent edition of "The Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations" and their amendments (United Kingdom).

SECTION 15: Regulatory Information**15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture**

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):

List entry in regulation: E1

List entry in regulation: H2

List entry in regulation: P8

This product may be subject to the Control of Major Accident Hazards Regulations (COMAH), and amendments if specific threshold tonnages are exceeded (United Kingdom).

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15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Ox. Sol. 2
Acute Tox. 3 (oral)
Eye Dam./Irrit. 2A
Aquatic Acute 1

M-factor acute: 1

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. Any other intended applications should be discussed with the manufacturer.

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Ox. Sol.	Oxidising solids
Acute Tox.	Acute toxicity
Eye Dam./Irrit.	Serious eye damage/eye irritation
Aquatic Acute	Hazardous to the aquatic environment - acute
H319	Causes serious eye irritation.
H301	Toxic if swallowed.
H400	Very toxic to aquatic life.
H272	May intensify fire; oxidizer.

Abbreviations

ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road.
ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

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The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.

Annex: Exposure Scenarios

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1. Formulation & (re)packing of substances and mixtures, (handling as solid), Industrial applications
SU3; SU3, SU9; ERC2; PROC4, PROC5, PROC8b, PROC9, PROC19, PROC26, PROC15
2. Formulation & (re)packing of substances and mixtures, (handling as solid in solution), (handling as melted mass), Industrial applications
SU3; SU3, SU10; ERC2; PROC4, PROC5, PROC8b, PROC9, PROC15, PROC19, PROC26
3. Formulation & (re)packing of substances and mixtures, (handling as solid), Professional applications
SU22; SU3, SU10; ERC2; PROC4, PROC5, PROC8b, PROC9, PROC15, PROC19, PROC26
4. Formulation & (re)packing of substances and mixtures, (handling as solid in solution), (handling as melted mass), Professional applications
SU22; SU3, SU8, SU9; ERC2; PROC3, PROC4, PROC8b, PROC9, PROC5, PROC15, PROC19, PROC26
5. Use as an intermediate, Use in chemical synthesis, Industrial applications
SU3; SU3, SU8, SU9; ERC6a; PROC3, PROC4, PROC8b, PROC15, PROC26
6. Use in Metallurgy, Use for Heat storage, Industrial applications
SU3; SU15; ERC7; PROC3, PROC4, PROC9, PROC25
7. Use as Corrosion inhibitor, (use in professional settings)
SU22; SU2b, SU3, SU17, SU22; ERC7; PROC5, PROC17, PROC20
8. Use in Metal surface treatment, (use in industrial settings), (handling as solid)
SU3; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC15
9. Use in Metal surface treatment, (handling as solid in solution), (handling as melted mass), (use in industrial settings)
SU3; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC15

1. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures, (handling as solid), Industrial applications
SU3; SU3, SU9; ERC2; PROC4, PROC5, PROC8b, PROC9, PROC19, PROC26, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). PROC15: Use a laboratory reagent.

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	Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
PROC4, PROC5	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
PROC9, PROC15	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.1 mg/m ³
Risk Characterization Ratio (RCR)	0.05
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week

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Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.1 mg/m ³
Risk Characterization Ratio (RCR)	0.05
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

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Contributing exposure scenario	
Use descriptors covered	PROC19: Hand-mixing with intimate contact and only PPE available. Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	

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Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation of preparations
Operational conditions	
Annual amount per site	700,000 kg
Minimum emission days per year Continuous	300
Emission factor air	0 %
Emission factor water	2 %
Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Treat wastewater (prior to discharge to STP) to provide the required removal efficiency of (%)	99 %
Wastewater treatment measures considered suitable are, e.g.	Oxidation
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	87.3 %
Total effc. of removal from wastewater after RMMs and STP(%)	87.3 %
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1
Maximum amount of safe use	4,257 kg/d
Risk from environmental exposure is driven by freshwater.	

2. Short title of exposure scenario

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Formulation & (re)packing of substances and mixtures, (handling as solid in solution), (handling as melted mass), Industrial applications

SU3; SU3, SU10; ERC2; PROC4, PROC5, PROC8b, PROC9, PROC15, PROC19, PROC26

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are	

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based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC15: Use a laboratory reagent. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	

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Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC19: Hand-mixing with intimate contact and only PPE available. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable chemically resistant gloves.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite

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	Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation of preparations
Operational conditions	
Annual amount per site	700,000 kg
Minimum emission days per year Continuous	300
Emission factor air	0 %
Emission factor water	2 %
Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	18,000 m ³ /d
Dilution factor river	10
Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Treat wastewater (prior to discharge to STP) to provide the required removal efficiency of (%)	99 %
Wastewater treatment measures considered suitable are, e.g.	Oxidation
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	87.3 %
Total effc. of removal from wastewater after RMMs and STP(%)	87.3 %
Assumed sewage treatment plant flow (m ³ /d)	2,000 m ³ /d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1

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Maximum amount of safe use	4,257 kg/d
Risk from environmental exposure is driven by freshwater.	

3. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures, (handling as solid), Professional applications
 SU22; SU3, SU10; ERC2; PROC4, PROC5, PROC8b, PROC9, PROC15, PROC19, PROC26

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). Use domain: professional
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 90 %
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
PROC4, PROC5	
Assessment method	ECETOC TRA v2.0 Worker; modified version, The concentration of the substance has been considered using a linear approach.
	Worker - inhalation, long-term - systemic
Exposure estimate	0.9 mg/m ³
Risk Characterization Ratio (RCR)	0.45
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	
Please note that a modified version has been used (see exposure estimates)	
Contributing exposure scenario	

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Use descriptors covered	PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC15: Use a laboratory reagent. PROC19: Hand-mixing with intimate contact and only PPE available. Use domain: professional
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
PROC8b, PROC9, PROC19	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the ECETOC TRA initial exposure value multiplied by a factor of 2.
PROC15	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.1 mg/m ³
Risk Characterization Ratio (RCR)	0.05
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	
Contributing exposure scenario	
Use descriptors covered	PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: professional covered by PROC8b covered by PROC9 covered by PROC5

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Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation of preparations
Operational conditions	
Annual amount per site	700,000 kg
Minimum emission days per year Continuous	300
Emission factor air	0 %
Emission factor water	2 %
Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Treat wastewater (prior to discharge to STP) to provide the required removal efficiency of (%)	99 %
Wastewater treatment measures considered suitable are, e.g.	Oxidation
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	87.3 %
Total eff. of removal from wastewater after RMMs and STP(%)	87.3 %
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1
Maximum amount of safe use	4,257 kg/d
Risk from environmental exposure is driven by freshwater.	

4. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures, (handling as solid in solution), (handling as melted mass), Professional applications

SU22; SU3, SU8, SU9; ERC2; PROC3, PROC4, PROC8b, PROC9, PROC5, PROC15, PROC19, PROC26

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where

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	opportunity for exposure arises. PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC15: Use a laboratory reagent. PROC19: Hand-mixing with intimate contact and only PPE available. PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: professional Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
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Operational conditions

Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass

Risk Management Measures

Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	

Exposure estimate and reference to its source

Assessment method	Qualitative assessment
	Worker - contact with eyes

Additional good practice advice

In case of possible exposure towards degradation products use a suitable respiratory protection
Avoid release of degradation products

Contributing exposure scenario

Use descriptors covered	ERC2: Formulation of preparations
Operational conditions	
Annual amount per site	700,000 kg
Minimum emission days per year Continuous	300
Emission factor air	0 %
Emission factor water	2 %
Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	18,000 m3/d

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Dilution factor river	10
Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Treat wastewater (prior to discharge to STP) to provide the required removal efficiency of (%)	99 %
Wastewater treatment measures considered suitable are, e.g.	Oxidation
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	87.3 %
Total effic. of removal from wastewater after RMMs and STP(%)	87.3 %
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1
Maximum amount of safe use	4,257 kg/d
Risk from environmental exposure is driven by freshwater.	

5. Short title of exposure scenario

Use as an intermediate, Use in chemical synthesis, Industrial applications

SU3; SU3, SU8, SU9; ERC6a; PROC3, PROC4, PROC8b, PROC15, PROC26

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC3: Use in closed batch process (synthesis or formulation). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	

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Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC15: Use a laboratory reagent. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	

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Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	ERC6a: Industrial use resulting in manufacture of another substance (use of intermediates)
Operational conditions	
Annual amount per site	8,000,000 kg
Minimum emission days per year Continuous	300
Emission factor air	0 %
Emission factor water	0.05 %

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Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	400,000 m3/d
Dilution factor river	40
Dilution factor coast	400
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	10,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1
Risk Characterization Ratio (RCR)	0.803
	Risk from environmental exposure is driven by freshwater.
Maximum amount of safe use	26,692 kg/d
Risk from environmental exposure is driven by freshwater.	

6. Short title of exposure scenario

Use in Metallurgy, Use for Heat storage, Industrial applications

SU3; SU15; ERC7; PROC3, PROC4, PROC9, PROC25

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC3: Use in closed batch process (synthesis or formulation). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	

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Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are	

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based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC25: Other hot work operations with metals Exposure is considered negligible.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
	Exposure is considered negligible.
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	ERC7: Industrial use of substances in closed systems.
Operational conditions	
Annual amount per site	1,500,000 kg
Minimum emission days per year Continuous	0
Emission factor air	0.00 %
Receive Surf. Water (Flow Rate).	18,000 m ³ /d
Dilution factor river	10
Dilution factor coast	100

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Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1

7. Short title of exposure scenario

Use as Corrosion inhibitor, (use in professional settings)

SU22; SU2b, SU3, SU17, SU22; ERC7; PROC5, PROC17, PROC20

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). PROC17: Lubrication at high energy conditions and in partly open process. PROC20: Heat and pressure transfer fluids in dispersive use but closed systems Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 10 %
Physical state	Solid in solution
Risk Management Measures	
Ensure segregation of worker from the source	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	
Contributing exposure scenario	
Use descriptors covered	ERC7: Industrial use of substances in closed systems. The environmental release is considered negligible.

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Operational conditions	
Annual amount per site	1,500,000 kg
Minimum emission days per year Continuous	0
Emission factor air	0.00 %
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10
Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1

8. Short title of exposure scenario

Use in Metal surface treatment, (use in industrial settings), (handling as solid)

SU3; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC1: Use in closed process, no likelihood of exposure. PROC3: Use in closed batch process (synthesis or formulation). PROC8a: Transfer of substance or preparation (charging/discharging) from/to ves-sels/large containers at non-dedicated facilities Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk	

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characterisation.	
Exposure estimate and reference to its source	
PROC1	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.01 mg/m ³
Risk Characterization Ratio (RCR)	0.005
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
PROC8a	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
PROC3	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.1 mg/m ³
Risk Characterization Ratio (RCR)	0.05
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). PROC15: Use a laboratory reagent. Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk	

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characterisation.	
Exposure estimate and reference to its source	
PROC4, PROC5	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
PROC9, PROC15	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.1 mg/m ³
Risk Characterization Ratio (RCR)	0.05
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.5 mg/m ³
Risk Characterization Ratio (RCR)	0.25
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment

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	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid – low dustiness
Duration and Frequency of activity	Application duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	ECETOC TRA v2.0 Worker
	Worker - inhalation, long-term - systemic
Exposure estimate	0.1 mg/m ³
Risk Characterization Ratio (RCR)	0.05
	The short-term exposure value corresponds to the long-term value multiplied by a factor of 2.
Assessment method	Qualitative assessment
	Worker - contact with eyes
Guidance to Downstream Users	
For scaling see: http://www.ecetoc.org/tra	

Contributing exposure scenario	
Use descriptors covered	ERC6b: Industrial use of reactive processing aids
Operational conditions	
	OECD ESD No.12 used, assessment independent from tonnage.
Minimum emission days per year Continuous	
Release to waste water from process	0.528 kg/d
Receive Surf. Water (Flow Rate).	18,000 m ³ /d
Dilution factor river	10

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Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	87.3 %
Total effic. of removal from wastewater after RMMs and STP(%)	87.3 %
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1
Risk Characterization Ratio (RCR)	0.794
Risk from environmental exposure is driven by freshwater.	

9. Short title of exposure scenario

Use in Metal surface treatment, (handling as solid in solution), (handling as melted mass), (use in industrial settings)

SU3; ERC6b; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC1: Use in closed process, no likelihood of exposure. PROC8a: Transfer of substance or preparation (charging/discharging) from/to ves-sels/large containers at non-dedicated facilities Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

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Contributing exposure scenario	
Use descriptors covered	PROC3: Use in closed batch process (synthesis or formulation). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes

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Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC15: Use a laboratory reagent. Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: $\geq 0\%$ - $\leq 100\%$
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are	

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based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). Use domain: industrial Solid in solution. For non-spraying processes (no aerosol generation), an inhalative exposure is considered to be not relevant.
Operational conditions	
Concentration of the substance	sodium nitrite Content: >= 0 % - <= 100 %
Physical state	Solid in solution, Melted mass
Risk Management Measures	
Ensure that no inhalable aerosols are generated.	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
In case of possible exposure towards degradation products use a suitable respiratory protection	
Avoid release of degradation products	

Contributing exposure scenario	
Use descriptors covered	ERC6b: Industrial use of reactive processing aids
Operational conditions	
	OECD ESD No.12 used, assessment independent from tonnage.
Minimum emission days per year Continuous	
Release to waste water from process	0.528 kg/d
Receive Surf. Water (Flow Rate).	18,000 m3/d
Dilution factor river	10

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Dilution factor coast	100
Other Factors: Environment	Indoor and outdoor use.
Risk Management Measures	
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	87.3 %
Total effic. of removal from wastewater after RMMs and STP(%)	87.3 %
Assumed sewage treatment plant flow (m3/d)	2,000 m3/d
Exposure estimate and reference to its source	
Assessment method	EUSES v2.1
Risk Characterization Ratio (RCR)	0.794
	Risk from environmental exposure is driven by freshwater.
